Modeling the seismic wave equation using a staggered grid finite-difference method optimized with a genetic algorithm

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Summary

Simulation of seismic waves is essential for subsurface imaging, where numerical dispersion remains a major challenge. Staggered grid finite-difference (SGFD) methods offer improved accuracy and stability, and this study applies a genetic algorithm (GA) to optimize SGFD coefficients for reduced dispersion in space and time. Numerical experiments on homogeneous and heterogeneous models show that GA-optimized coefficients significantly decrease dispersion, even with lower-order approximations, thereby reducing computational cost. For instance, optimizing 12th-order coefficients required only ~20 s on a standard 64 GB RAM system. This work reproduces the study "Modeling the seismic wave equation using a staggered grid finite-difference method optimized with a genetic algorithm" by Mounika Vanga and Maheswar Ojha and extends it by further strengthening the GA algorithm and also testing the GA on the lens model.

Introduction

Precise seismic wave simulation is crucial for subsurface imaging; nevertheless, numerical dispersion persists in constraining the efficacy of finite-difference (FD) approaches. Traditional finite difference methods are prevalent due to their simplicity; however, they distort phase velocities when grid spacing and time intervals are insufficient (Kelly et al., 1976). Staggered grid finite-difference (SGFD) approaches enhance accuracy and stability by the use of first-order stress–strain equations, hence minimizing interpolation errors and improving convergence (Igel et al., 1992).

To further reduce dispersion, several studies have explored optimization-based FD schemes rather than relying solely on Taylor-series formulations (Finkelstein & Kastner, 2006; Liu & Sen, 2011). A major advance was presented by Vanga & Ojha (2025), who used a genetic algorithm (GA) to optimize SGFD coefficients based on space–time dispersion relations. Their results demonstrated broader wavenumber coverage and significantly reduced numerical errors compared to conventional and time–space FD methods.

In this work, we reproduce the SGFD–GA methodology of Vanga & Ojha (2025) and extend it by further strengthening the genetic algorithm performance and also testing the GA on the lens model.

Methodology

Seismic wave propagation is modeled using the acoustic wave equation:

$$\frac{\partial^2 u(\mathbf{x},t)}{\partial t^2} = v^2(\mathbf{x})\nabla^2 u(\mathbf{x},t) + s(t),$$

Where; $u(\mathbf{x},t)$ is the pressure wavefield, $v(\mathbf{x})$ is the P-wave velocity, and $s(t)$ is the source term. The equation is solved numerically using a staggered-grid finite-difference (SGFD) scheme, in which spatial and temporal derivatives are evaluated at staggered locations to improve numerical accuracy and stability.

Time derivatives are approximated using a second-order central difference, while spatial derivatives are computed using a high-order staggered finite-difference operator expressed as

$$\frac{\partial u}{\partial x} \approx \sum_{i=1}^{M} a_i \left(u_{x+i-\frac{1}{2}} - u_{x-i+\frac{1}{2}} \right),$$

Where; a_i are the finite-difference coefficients and M controls the order of spatial accuracy. This formulation reduces interpolation errors and improves phase accuracy compared to conventional collocated-grid schemes.

Numerical dispersion is analyzed by substituting a plane-wave solution,

$$u = e^{i(kx-\omega t)},$$

into the discrete wave equation, yielding the numerical dispersion relation. Dispersion is quantified using the phase-velocity ratio

$$\delta(kh) = \frac{v_{\text{num}}}{v_{\text{true}}},$$

Where; k is the wavenumber and h is the grid spacing. An ideal numerical scheme satisfies $\delta = 1$ over the wavenumber range of interest. Because dispersion depends on both spatial and temporal discretization, optimization based solely on spatial accuracy is insufficient.

To minimize numerical dispersion, the SGFD coefficients are optimized using a genetic algorithm (GA). The GA treats the finite-difference coefficients a_i as optimization variables and evolves them through selection, crossover, and mutation (Figure 1). The fitness function is defined as:

$$\text{Fitness} = w_1\,\text{err}_{\text{mean}} + w_2\,\text{err}_{\text{std}},$$

where err_{mean} and err_{std} are the mean and standard deviation of the dispersion error computed from the phase-velocity ratio, and $w_1 + w_2 = 1$. Additional weighting is applied to low-wavenumber components to preserve long-wavelength accuracy.

The optimized coefficients are applied to seismic wave simulations in homogeneous and heterogeneous velocity

models. For heterogeneous media, coefficients optimized for the minimum velocity are used to control dispersion across the entire velocity range. Synthetic wavefields and shot gathers are generated and compared with high-order reference solutions to assess dispersion reduction, stability, and computational efficiency.

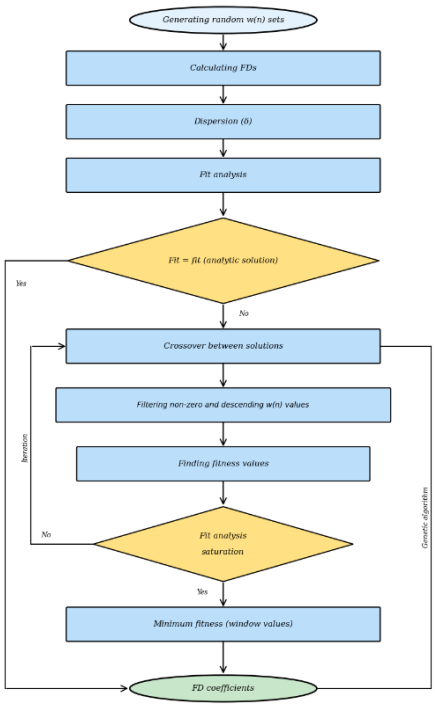

Figure 1: Workflow for optimization of finite-difference (FD) coefficients

Results

The dispersion characteristics of the proposed genetic-algorithm (GA)-optimized staggered-grid finite-difference (SGFD) scheme was first evaluated using phase-velocity ratio analysis. Compared to conventional and time–space staggered-grid methods, the proposed approach exhibits a substantial reduction in numerical dispersion over a broad range of wavenumbers (Figure 2). Notably, lower-order GA-optimized SGFD schemes achieve dispersion performance comparable to, or better than, much higher-order conventional and time–space schemes, demonstrating improved accuracy without increasing computational cost.

A direct comparison between GA-optimized staggered-grid and GA-optimized explicit finite-difference formulations further highlights the advantage of the staggered-grid approach. As shown in Figure 3, the GA-based SGFD method provides broader wavenumber coverage and reduced dispersion relative to the GA-based explicit finite-difference method, confirming that the staggered-grid formulation plays a critical role in improving numerical accuracy in addition to coefficient optimization.

The stability characteristics of the proposed method were further examined by comparing the stability factor as a function of approximation order (Figure 4). The results show that the GA-optimized staggered-grid scheme consistently exhibits higher stability than the explicit finite-difference method and maintains comparable or improved stability relative to conventional and time–space staggered-grid approaches. Notably, the staggered-grid formulation allows for larger stable time steps across increasing approximation orders, confirming that the proposed GA-based SGFD method enhances stability without sacrificing numerical accuracy.

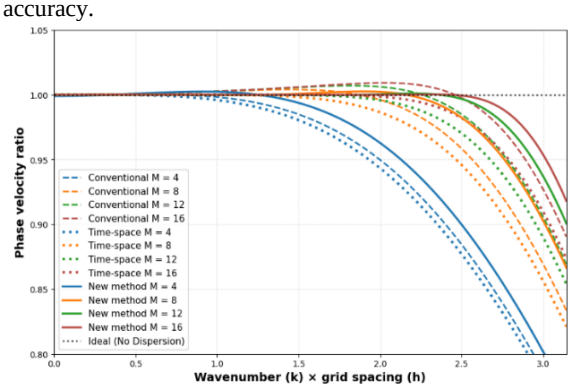

Figure 2: Plot of phase velocity ratio (dispersion) versus the product of the wavenumber (k) and grid spacing (h) for conventional, time-space and new genetic algorithm-based staggered grid methods, across various orders (M) of approximations.

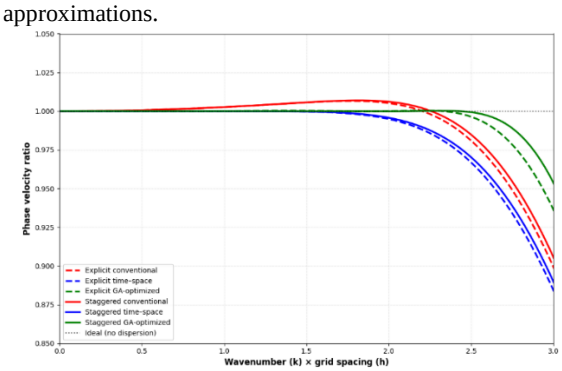

Figure 3: Dispersion curves versus kh for conventional, time-space, and new genetic algorithm (GA)-based staggered grid finite-difference methods (FDM) compared with the GA-based explicit FDM for a 12th-order approximation.

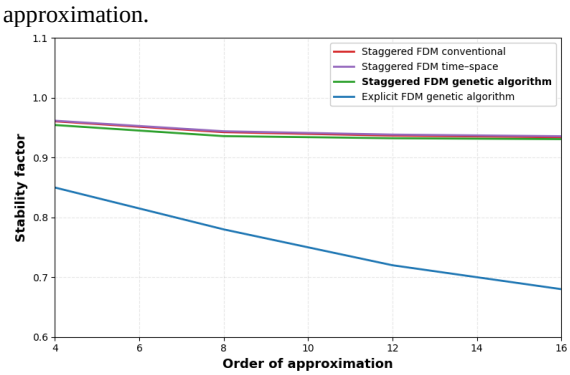

Figure 4: Comparison of stability conditions for the proposed genetic algorithm (GA)-based staggered grid finite-difference method (FDM; green) with the GA-based explicit FDM (blue), conventional staggered grid FDM (red), and time-space staggered grid FDM (purple).

Numerical examples of 2D wave propagation

The impact of reduced numerical dispersion is further illustrated through two-dimensional wavefield simulations in a homogeneous velocity model. Wavefield snapshots generated using the proposed GA-optimized SGFD method show cleaner, more coherent wavefronts and significantly reduced phase errors compared to conventional and time–space schemes, particularly at lower approximation orders (Figure 5). These results demonstrate that the improvements observed in the dispersion analysis directly translate into more accurate and physically consistent wavefield propagation.

To further evaluate the performance of the proposed method in complex settings, additional simulations were conducted for heterogeneous velocity models, including a horizontal-layered model, a salt-dome model, and a lens model. In the horizontal-layered model, the GA-optimized SGFD scheme produces wavefields with improved event continuity and reduced dispersion compared to the conventional staggered-grid method, closely matching the high-order reference solution (Figure 6). For the salt-dome model, the proposed method demonstrates superior wavefield quality with reduced numerical artifacts and enhanced stability in regions of strong velocity contrast (Figure 7). In the lens model, the GA-optimized scheme accurately captures wave refraction and focusing effects caused by the localized velocity anomaly, while significantly reducing dispersion relative to the conventional method (Figure 8). These results confirm that the GA-optimized SGFD approach remains robust and effective for realistic geological models.

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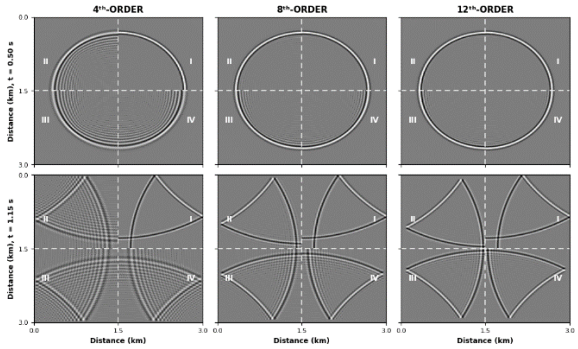

Figure 5: Snapshots of 2D wave propagation in a homogeneous model for 4th, 8th, and 12th orders. Quarter I show the reference wavefield generated using the 40th-order conventional method; quarter II is the new genetic algorithm-based method; quarter III is the time-space method; and quarter IV is the conventional method.

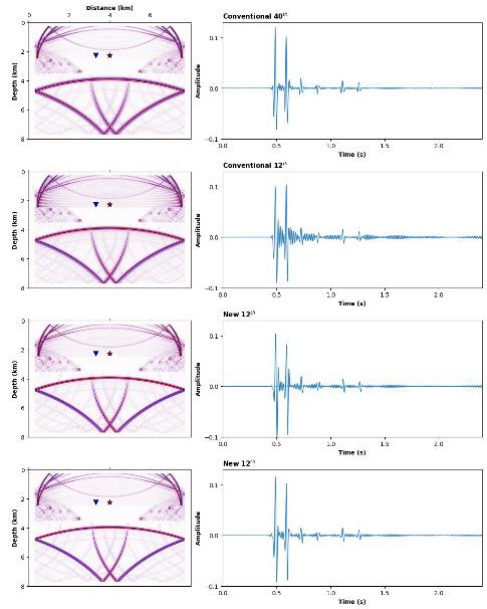

Figure 6: Acoustic wavefield simulations for horizontal-layered model (Figure 8A). (A) Reference solution computed using the 40th-order conventional method. (B) 12th-order conventional method. (C) 12th-order GA-optimized staggered-grid method with coefficients optimized for the minimum velocity. (D) 12th-order GA-optimized staggered-grid method with velocity-dependent coefficients. Source and receiver locations are indicated by red stars and blue triangles, respectively.

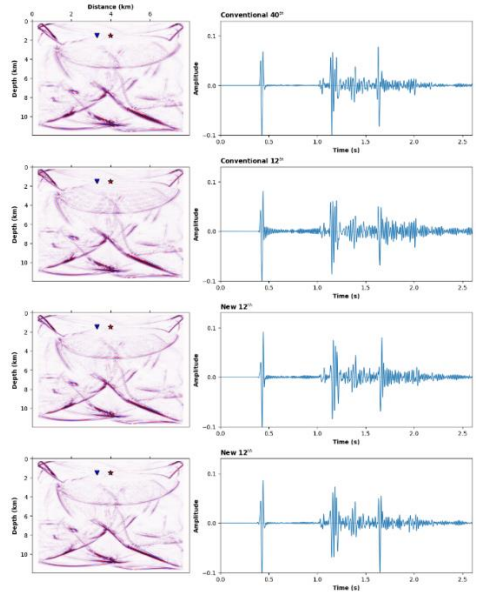

Figure 7. Acoustic wavefield simulations for the BP salt-dome model (Figure 8B). (A) Reference solution computed using the 40th-order conventional method. (B) 12th-order conventional method. (C) 12th-order GA-optimized staggered-grid method with coefficients optimized for the minimum velocity. (D) 12th-order GA-optimized staggered-grid method with velocity-dependent coefficients. Source and receiver locations are indicated by red stars and blue triangles, respectively.

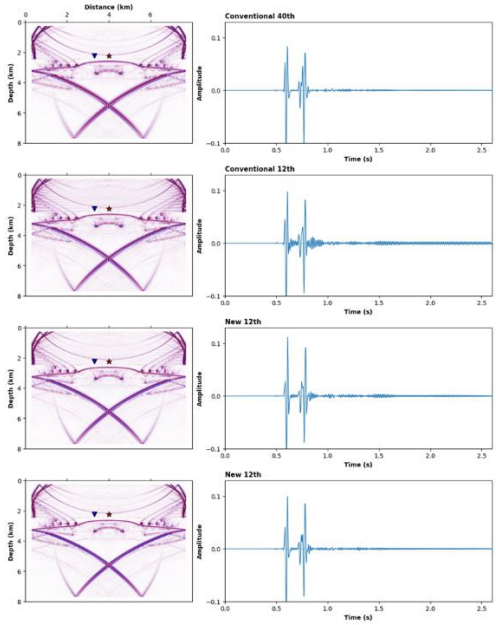

Figure 8. Acoustic wavefield simulations for the lens model (Figure 8B). (A) Reference solution computed using the 40th-order conventional method. (B) 12th-order conventional method. (C) 12th-order GA-optimized staggered-grid method with coefficients optimized for the minimum velocity. (D) 12th-order GA-optimized staggered-grid method with velocity-dependent coefficients. Source and receiver locations are indicated by red stars and blue triangles, respectively.

Conclusion

We presented a genetic-algorithm–optimized staggered-grid finite-difference (SGFD) scheme for seismic wavefield modeling that significantly reduces numerical dispersion while maintaining computational efficiency. By optimizing finite-difference coefficients using a dispersion relation that accounts for both spatial and temporal discretization, the proposed method achieves high accuracy over a broad wavenumber range without requiring high-order approximations.

Dispersion analysis and wavefield simulations demonstrate that the GA-optimized SGFD scheme outperforms conventional and time–space staggered-grid methods, as well as GA-optimized explicit finite-difference formulations. The reduced dispersion observed in the phase-velocity analysis translates directly into cleaner, more coherent wavefronts in physical space, even at lower approximation orders.

The proposed approach provides an efficient and robust framework for accurate seismic wave propagation modeling, with particular relevance for applications requiring high-resolution imaging and large-scale simulations. The ability to achieve improved accuracy using lower-order schemes makes the method especially attractive for computationally demanding problems such as full waveform inversion and complex subsurface modeling.

Key words

Numerical dispersion; Seismic wavefield; Staggered grid finite-difference method; Genetic algorithm; Modeling; Optimization

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